

WON KO

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SUMMARY

Senior Researcher in autonomous robotics specializing in multi-modal sensor fusion (camera/LiDAR/IMU), sensor calibration and time synchronization, probabilistic state estimation (3D SLAM / ESKF), and 3D perception (detection/segmentation/keypoints/depth) for reliable real-world autonomy and quantitative measurement validation.

TECHNICAL SKILLS

Languages	Embedded C/ C++, Python (OpenCV, TensorFlow, PyTorch)
Software	Isaac SIM, Git, Docker, MATLAB/Simulink, V-REP, Solidworks.
Operating Systems	ROS/ROS 2, Unix/Linux system (FreeBSD, Ubuntu, Debian), Windows

EDUCATION

University of California, Santa Cruz Master of Science in Computer Engineering Emphasis in Robotics and Control MS, "Autonomously Driving Tricycle with Monocular Vision" Advisor: Prof. Dejan Milutinović	August 2022
University of California, Santa Cruz Bachelor of Science in Computer Engineering Concentration on Robotics and Control	December 2019

RESEARCH EXPERIENCE

Senior Researcher / Future Tech R&D Lab, Mobyus	November 2024 - Present
- Research and develop autonomy and perception for AMR/AFL platforms: multi-modal sensor fusion, state estimation (3D SLAM), and motion planning/control on real hardware. - Developed and validated 3D SLAM pipelines (LiDAR–inertial odometry with ESKF/Kalman filtering; visual–inertial odometry); achieved under 30 mm localization error on testbeds and improved repeatability in field trials. - Built multi-sensor 3D perception for obstacle detection and scene understanding (camera + LiDAR fusion), including the time synchronization and latency-aware integration for safe navigation. - Designed calibration pipelines using least-squares optimization: LiDAR–IMU and camera–IMU extrinsic calibration; synchronized multi-sensor data capture and quantitative evaluation (RMS error, latency, robustness). - Productionized and deployed autonomy modules with reliability features (startup checks, recovery behaviors, E-stop/watchdogs), PLC workflow integration, and logging/telemetry; maintained build/release workflows (colcon, containers, CI/CD).	
Researcher / Robotics and Control Lab, UC Santa Cruz	September 2020 - August 2023
- Researched probabilistic state estimation and stochastic nonlinear control for GPS-free navigation: Kalman-filter-based sensor fusion, system identification, and robustness under real sensor noise and timing constraints. - Implemented calibration/estimation experiments using multi-view geometry and least-squares optimization; created synchronized data collection and quantitative evaluation workflows (RMS error, latency, repeatability).	

TEACHING EXPERIENCE

Teaching Assistant / University of California, Santa Cruz

Mechatronics (ECE 118)

Spring 2022, Fall 2021

Santa Cruz, CA

- Led lab sections on event-driven programming, hierarchical state machines (HSMs), sensor interfacing, and end-to-end system integration; provided debugging support and graded coursework with the TA team.
- Maintained and repaired lab hardware; performed safety checks and managed spare parts/inventory to minimize downtime.

Teaching Assistant / University of California, Santa Cruz

Computer Architecture (CSE 120)

Spring 2021, Winter 2022

Santa Cruz, CA

- Led RISC-V and hardware design labs using Verilog/SystemVerilog; guided students through simulation, debugging, and design verification.
- Held weekly office hours and graded assignments/exams; supported performance analysis and correctness debugging for lab implementations.

PROJECTS

Autonomous Forklifts (AFL)

November 2024 - October 2025

- Implemented motion planning/control for autonomous forklifts: Frenet-frame trajectory generation and tracking with B-spline smoothing for obstacle avoidance and repeatable docking; deployed on three-wheeled (SD-type motor) platforms.
- Built and validated a LiDAR-IMU localization stack (ESKF-based LIO / 3D SLAM) with quantitative testbed evaluation (e.g., ≤ 30 mm position error) and field repeatability checks on real hardware.
- Developed multi-modal 3D perception for safety: depth camera (RealSense D435f) + 2D LiDAR fusion for obstacle detection; improved robustness under occlusion/reflective surfaces with latency-aware integration.
- Developed precision keypoint perception for pallet handling using SICK Visionary-T Mini: Transformer-based fork-pocket (hole-center) keypoint detection and pallet sizing with CAD-geometry constraints; achieved hole detection at 4 m with 0.5 deg angular error for insertion/alignment.
- Built stereo-based free-space estimation using ZED2i for truck/trailer loading; integrated 3D scene understanding with object detection (YOLOv12, RF-DETR) to estimate available placement volume and feasibility.

Continental Intelligent Intersection – *co-sponsored by Continental AG*

September 2018 - June 2019

- Developed and evaluated ML models (CNN, RNN, SVM, Hidden Markov Model) for traffic-scene understanding and risk prediction at intersections.
- Customized YOLOv3 for project-specific object detection and trained on curated datasets from heterogeneous sensors (short/long-range radar and camera).
- Built synchronized multi-sensor data capture and evaluation metrics (RMS error, latency, robustness) for lab/field validation.
- Built a data pipeline for sensor synchronization/labeling and deployed inference to detect safety-critical scenarios (e.g., pedestrian crossings, left-turn conflicts) using Python + OpenCV.